

Transcendental Avatar: Experiencing Bioresponsive Avatar of the Self for Improved Cognition

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Figure 1: Transcendental Avatar, on three different background with two conditions - a high particle-points avatar, and a low particle-lines

ABSTRACT

Biofeedback is a well-known form of therapy in which the patient receives sensory and auditory input on their physiology to help them reflect on, recognize, and be aware of their own state in order to improve their cognitive and emotional functioning. In our approach, a user's self-avatar is generated based on their physiological state, specifically their heart rate and electrodermal activity. Moreover, based on self-reported anxiety symptoms, we investigate if the biofeedback self-immersive avatar can help reduce stress levels.

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1 INTRODUCTION

A person's mental health affects their entire life. In order to participate fully in social life, recognize our needs, attend social events, work more productively and manage various situations, we need to be mentally healthy and stress-free. Anxiety and depression are the most common mental illnesses in the world, affecting over 264 million people of all ages, according to a scientific brief published by the World Health Organization (WHO) on March 2, 2022¹. Therapy sessions are therefore a necessary form of treatment. Biofeedback is an example of a treatment method that re-experiences a patient's physiological state. As patients try to regulate their physiological signals, they are typically required to examine raw physiological signals [Yu et al. 2018]. Previous works have explored immersive biofeedback for a variety of applications using VR, a platform that allows users to become fully immersed in an alternative environment. Researchers have looked into its applications for agility transfer [Hajika et al. 2019], multi-sensory feedback [Morat et al. 2021], and other things. This data is not portrayed in these works as the avatar of the self, though, and this is a crucial factor to take into account when developing biofeedback treatment for self-reflection. In order to achieve this, we suggest Transcendental Avatar, a VR experience in which the user's avatar is portrayed as a reflection of their

¹https://www.who.int/publications/i/item/WHO-2019-nCoV-Sci_Brief-Mental_health-2022.1

emotional and cognitive state. Emotional Beasts, a related project, investigated the possibility of using emotionally adaptive avatars to display users' physiological conditions [Bernal and Maes 2017]. Although we see our work as a self-help tool, the authors primarily concentrated on self-expression. This work's contributions are:

- To test the effectiveness of this technique as a supplemental tool to reduce anxiety and stress levels, we develop a virtual reality system called "Transcendental Avatar."
- We go over how using a self-avatar for biofeedback might help with relaxing.



Figure 2: The Transcendental Avatar system

2 DESIGN AND IMPLEMENTATION

The user's avatar and the emotional stimuli impact how the Transcendental Avatar VR experience is designed. We were inspired by personification's abstract form when creating the user avatar. The user's point cloud is captured using the Azure Kinect2 sensor, then using Unity3D for creating a VR environment, and adjustments are made to the user's abstract representation. For instance, the user's self-avatar appears more abstract and less vivid when the emotional state is low, and the mental workload is high. We design a biofeedback avatar that recognizes and reacts to the user's emotional and cognitive condition. In other words, the avatar must receive bioresponsive information from the user's physiology. Heart rate variability (HRV) and electrodermal activity (EDA) are the two key physiological signals that we detect to do this. It has been demonstrated that these signals reflect cognitive load [Gupta et al. 2019] and emotional [Gupta et al. 2021] burden of the user. To gather the required information, we developed a customized sensing device based on the ESP32 that was worn on the left hand. Plethysmograph (PPG) is used to measure HRV, with a sampling rate of 50Hz, as opposed to the EDA signal, which was collected at 4.545Hz. These signals are sent to a program, where signal processing is carried out. An increase in perceived cognitive load causes both the parasympathetic and sympathetic components of the HRV signal to increase, which is reflected in a rise in HRV [Solhjoo et al. 2019]. We observe the phasic element of EDA, which is inversely correlated with emotional arousal [Caruelle et al. 2019].

3 PLANNED USER STUDY

We primarily concentrate on the presence of the biofeedback self-avatar in the VR system and its impact on relaxation or emotional understanding for the design of our experiment. Three different

avatar designs—a high particle-points avatar, a medium particle-squares avatar, and a low particle-lines avatar—are used in 10 other natural surroundings as part of the experiment. In our experimental design, the user can freely perceive the changes in the avatar based on their mental state or interact with the displayed avatar's presence and appearance by concentrating on and controlling their mental and emotional state.

4 CONCLUSION

Transcendental Avatar is a VR approach that personifies the user as an abstract avatar representing their emotional and cognitive condition. Our autonomous system intends to encourage soothing and relaxing techniques. After reviewing existing research on combining virtual reality with calming methods, Transcendental Avatar was formed. In addition, our objective is to develop a novel relaxing technique with virtual reality technology in the context of the worsening global situation regarding mental health issues. Global concerns like stress and anxiety demand more technological methods and therapies to deal with them. To encourage self-relaxing and relaxation practices, we wish to offer a VR system, Transcendental Avatar - a personification of users based on physiological signals. The potential outcome of this study is the presentation of a technique that can help people decompress and reduce stress.

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